RESEARCH PROJECT PROPOSAL

Proxy Models for Needle-in-a-Haystack Semantic Filters

Team: EmbedSeek

Project Members

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Abstract

Semantic filters evaluate natural-language predicates over large text collections, but labeling every record with a large language model (LLM) is expensive. Lightweight proxies reduce inference cost, yet a small uniformly sampled training set may contain almost no positives when the target class is rare. This project studies *Direction B: finding diverse positives cheaply* using multiple pretrained text-embedding representations. We hypothesize that different embeddings expose complementary neighborhoods and that a sampling policy can learn, from limited labeling feedback, which representation is most productive for a given predicate. We will first compare fixed embedding spaces using PCA/UMAP visualizations and quantitative neighborhood diagnostics, then implement query-based, positive-anchor, and diversity-aware samplers. Our principal proposed method is a lightweight multi-armed bandit that adaptively allocates a fixed LLM labeling budget across embedding spaces. We will evaluate positive yield, independently measured positive diversity, downstream proxy F1 on human gold labels, and total computational cost. The goal is to identify when adaptive representation selection improves on strong single-embedding baselines, and when its added overhead is not worthwhile.

1 Motivation and Research Gap

Modern semantic filters evaluate predicates such as `AI.IF('contains a credible threat', comment)` using an LLM. On large tables this is slow and expensive. Chung et al. [1] propose labeling a small sample and fitting a lightweight proxy model on text embeddings. However, when positives are rare, random sampling often produces too few examples to learn a useful decision boundary; for example, at 0.1% prevalence, 500 random draws contain only 0.5 positives in expectation. Reweighting cannot recover positive examples that were never encountered.

Barbosa et al. [2] instead find rare positives through cluster-based adaptive sampling, but repeated sampling and retraining may impose substantial overhead. Related approaches such as SEALS [5] and AnchorAL [6] exploit neighbors around informative labeled examples. We investigate an additional, practical variable: *the embedding representation itself*. General-purpose sentence embeddings, retrieval-tuned embeddings, and representations adapted to a domain may rank the same items differently. A strategy that always samples from a single representation may miss entire types of positives.

Research gap. Can an efficient sampling policy use a small amount of LLM feedback to allocate queries among *multiple fixed pretrained embedding spaces*, discovering more diverse positives than random or single-embedding methods, without losing the savings to embedding and selection overhead?

2 Research Questions and Hypotheses

RQ1. Representation: How much do rare-positive neighborhoods, query rankings, and local purity vary across pretrained embeddings for the same predicate?

RQ2. Discovery: Can query-similarity, positive-anchor search, and diversity-aware sampling find more positive examples and distinct positive subtypes per LLM label than random selection?

RQ3. Adaptation: Does a feedback-driven policy that chooses among embedding spaces outperform the best prespecified single-embedding strategy at a fixed label budget?

RQ4. Downstream value and cost: Do improved samples yield higher held-out proxy F1, and at what computational or monetary cost?

Primary hypothesis. Different pretrained embedding spaces capture complementary neighborhoods; an adaptive selection policy can exploit this complementarity to discover more *diverse* positives under a limited labeling budget. This is a testable hypothesis, not an assumed outcome. The method may fail if all embeddings return nearly identical neighbors, feedback is too sparse, or representation-computation costs dominate.

3 Proposed Approach

3.1 Multiple Embedding Spaces and Exploratory Analysis

We will begin with two or three off-the-shelf encoders from different representation families (for example, a general-purpose Sentence Transformer and a retrieval-oriented text encoder), choosing concrete checkpoints after a Week 1 feasibility check. No encoder will be fine-tuned in the main study. For model m, define normalized record and query vectors

$$z_i^{(m)} = \frac{f_m(x_i)}{\|f_m(x_i)\|_2}, \qquad z_q^{(m)} = \frac{f_m(q)}{\|f_m(q)\|_2}.$$

Each embedding model has its *own* dimensionality and similarity scale; comparisons will use within-model normalized similarity or ranks rather than assuming coordinates are interchangeable.

PCA and UMAP will visualize labeled positives and negatives in each space, but *only as exploratory diagnostics*. Sampling, nearest-neighbor search, and final proxy learning will use original high-dimensional representations. We will also measure top-k positive enrichment, neighborhood purity, overlap of retrieved sets between encoders, and the distribution of positives across clusters. Ground-truth labels used to prepare retrospective diagnostic plots will never be available to the sampler during an experimental run.

3.2 Fixed-Representation Sampling Baselines

At identical acquisition budgets $B \in \{100, 250, 500, 1000\}$, we will compare the following samplers:

(a) **Uniform random sampling:** the primary baseline.

(b) **Query-seeded retrieval:** rank items by cosine similarity between a predicate embedding and record embeddings in each model separately.

(c) **Diversity/cluster coverage:** select from multiple embedding-space regions to avoid near-duplicate positives and improve coverage.

(d) **Positive-anchor search:** after finding positives, retrieve previously unlabeled neighbors of these anchors, inspired by SEALS and AnchorAL; include an exploration fraction to avoid local trapping.

These experiments isolate the effects of *embedding choice* and *sampling rule* before attempting adaptation. We will retain one strong fixed-embedding method as the principal competitive baseline, not compare only with random sampling.

3.3 Proposed Method: Adaptive Multi-Embedding Sampling

Our primary algorithm will treat each pretrained embedding space as an arm of a small multi-armed bandit. At each round, the policy selects one representation, proposes a batch of unlabeled records using query similarity and/or positive-anchor retrieval, obtains LLM labels, and updates its allocation for subsequent rounds. Thompson sampling or an upper-confidence-bound policy will be considered; the initial implementation will use a simple interpretable reward and a limited number of rounds to control latency.

For representation m at round t, a candidate reward is

$$r_{m,t} = \frac{p_{m,t} + \lambda d_{m,t}}{b_{m,t}},$$

where $p_{m,t}$ is newly discovered positives, $d_{m,t}$ is a *proxy for novel positive types* (estimated from a common diversity space or an independent grouping), $b_{m,t}$ is the number of labels requested, and λ controls the diversity preference. The same sampled item will never be labeled twice. We will test reward variants to determine whether a simple positive-yield reward already suffices.

Critical evaluation safeguard. We will not define final positive-subtype diversity solely by the clusters of the embedding selected by a sampler, since that would reward each method according to its own representation. Where datasets provide meaningful subtype labels, we will use them; otherwise we will

report diversity under a shared, fixed representation with manual inspection of representative cases. We will explicitly distinguish this imperfect proxy from verified semantic subtype coverage.

3.4 Optional Extension: Combining Representations

If the core method and comparisons are complete, we will investigate a learned combination of within-model query-similarity scores:

$$S(x_i, q) = \sum_{m=1}^{M} w_m \, \mathrm{sim}(z_i^{(m)}, z_q^{(m)}), \quad w_m \geq 0, \quad \sum_m w_m = 1.$$

Weights could be adapted from labeled examples, but this is *not required* for the core contribution: extremely few positives may make weight fitting unstable. End-to-end encoder training is out of scope.

3.5 Proxy Training and Labeling Protocol

For every sampler, we will train the *same* class-weighted logistic-regression proxy using a fixed reference embedding representation, and choose its classification threshold on validation data. Fixing the downstream features and model architecture isolates the effect of *which records were selected* instead of conflating sampling with classifier capacity. As a secondary ablation, we may evaluate a proxy trained on concatenated or combined features, applied consistently across all samplers.

We will first run repeatable simulations using existing human annotations as a *labeling oracle*; labels are hidden until a sampler requests an item. Next, for a bounded subset of settings, we will call an actual LLM to label selected samples and quantify label noise and selection effectiveness under real labeling costs. All final precision, recall, and F1 measurements use a separate, human-labeled test set. Train/pool, validation, and test partitions will be disjoint, with no test-label access during sampler development. Validation prevalence and threshold selection will be handled carefully for extreme imbalance.

4 Datasets and Experimental Design

Three complementary datasets will be selected after a brief feasibility audit:

- **MTEB classification data:** establish a reproducible proxy baseline and construct controlled sampling pools with 20%, 5%, 1%, 0.5%, and 0.1% positives.
- **Jigsaw Toxic Comments:** investigate naturally rare classes such as *threat* and *identity hate*, for which human annotations are available.
- **BEIR (e.g., SciFact or FiQA):** evaluate nuanced query–document relevance, where positives may be extremely sparse; respect the official query/document relevance judgments and avoid treating unjudged records automatically as true negatives.

Train/pool, validation, and test partitions will be established *before* class-prevalence manipulation, and repeated sampling will use matched budgets and random seeds (target: 10 seeds for simulated-label experiments). Each dataset/predicate will share the same candidate pool and label oracle across samplers. For BEIR, evaluation will be limited to tasks where judged positives and defensible negatives permit reliable comparisons; otherwise it will be explicitly reported as an acquisition-only case. We will report exact positive counts and uncertainty intervals, especially at 0.1% prevalence.

5 Evaluation and Expected Evidence

Primary acquisition metrics: number of positives found per 100 labels; cumulative positives versus budget; and independent positive-subtype coverage when reliable subtype definitions exist. For adaptive methods, plot reward and embedding-arm allocation over labeling rounds.

Downstream proxy metrics: F1, precision, recall, and average precision on held-out human labels. The classifier and threshold-selection protocol will be identical across sampling methods. The key comparison is not simply whether adaptive selection beats random sampling, but whether it beats a *strong fixed single-embedding* baseline.

Costs: measure per-encoder embedding and index build time, query ranking, clustering, adaptation updates, LLM labeling, classifier training, and table-wide inference. Distinguish cold-start from cached-embedding scenarios:

$$C_{\text{adaptive}} = \sum_{m=1}^{M} C_{\text{embed},m} + C_{\text{index}} + C_{\text{sample}} + Bc_{\text{LLM}} + C_{\text{train}} + C_{\text{infer}}.$$

We will identify when extra representations pay for themselves in discovery efficiency and downstream proxy quality. For a fixed label budget, extra embeddings need not reduce monetary labeling cost; any claim of total cost savings must be demonstrated rather than assumed.

Planned figures: (1) side-by-side PCA/UMAP views and high-dimensional neighborhood statistics for different embeddings; (2) positives and subtype coverage versus labels; (3) adaptive representation allocations across rounds; (4) test F1 versus budget; and (5) performance versus end-to-end cost. Visual separation in 2D will not be treated as evidence of classifier separability.

6 Milestones: October 11–December 10, 2026

- **Weeks 1–2 (Oct 11–24): Understanding the problem and datasets.** Read the main papers and related work to better understand existing approaches to proxy models and rare-positive sampling. Explore the proposed datasets, examine class distributions, and become familiar with the available labels and semantic filtering tasks.
- **Weeks 3–4 (Oct 25–Nov 7): Exploring embedding representations.** Generate embeddings using different pretrained models and use PCA, UMAP, and clustering to investigate how the data is represented. Study whether positive examples form meaningful neighborhoods and begin designing experiments to compare the representations and identify promising sampling strategies.
- **Week 5 (Nov 8–14): Evaluating and refining our proposed approach.** Based on our initial experiments and analysis of the embedding spaces, assess whether the proposed multi-armed bandit approach is appropriate for the problem. Investigate its potential advantages and limitations, and determine whether to proceed with the original design, refine it, or explore alternative strategies suggested by our findings.
- **Weeks 6–7 (Nov 15–28): Implementation and experimentation.** Implement the selected sampling strategies and compare their ability to discover rare and diverse positive examples under fixed labeling budgets. Evaluate their impact on downstream proxy performance, investigate unsuccessful approaches, and refine the methods based on experimental findings.
- **Week 8 (Nov 29–Dec 5): Evaluating results and preparing deliverables.** Consolidate experimental results, compare successful and unsuccessful approaches, and analyze performance and computational costs. Prepare figures and begin writing the final project report, focusing on what we learned from the experiments.
- **Final days (Dec 6–10): Report, presentation, and submission.** Finalize the project report with our methodology, findings, limitations, and conclusions. Prepare a presentation discussing what worked, what did not work, and possible future directions. Submit the final deliverables by December 10.

7 Scope, Risks, and Success Criteria

The **minimum viable study** compares several pretrained embeddings and fixed sampling rules against uniform random sampling, measuring both positive yield and downstream proxy quality. The **primary novel component** is a lightweight bandit policy that allocates sampling among embeddings using labeling feedback. Learning continuous embedding-combination weights is optional; training new embedding foundation models or building a production database engine is out of scope.

Main risks include embeddings with overlapping retrieval results, extreme scarcity of feedback, misleading 2D projections, biased or unverified diversity measures, noisy LLM labels, and prohibitive multi-encoder overhead. We will report these limitations explicitly. Success is a reproducible, well-controlled answer about *when adaptive representation choice is useful*, supported by positive-yield, diversity, held-out F1, and cost comparisons. A carefully characterized negative result remains valuable.

8 Final Deliverables

The final submission will include the following:

- **Project report:** A written report presenting our research questions, experimental setup, methods, quantitative findings, and analysis. The report will discuss which embedding representations and sampling strategies worked, which did not, the reasons suggested by our experiments, and the limitations of our conclusions.
- **Final presentation:** A concise presentation summarizing the motivation, methods, principal results, and lessons learned. It will explicitly compare successful and unsuccessful approaches, including negative results and trade-offs in labeling efficiency, diversity, proxy F1, and computational cost.
- **Reproducible experimental artifacts:** Dataset splits, evaluation scripts, implementations of baseline and adaptive samplers, experiment configurations, and figures supporting the report and presentation.

References

[1] Y. Chung et al. "100x Cost & Latency Reduction: Performance Analysis of AI Query Approximation using Lightweight Proxy Models." *Proceedings of the ACM on Management of Data*, 2026. doi:10.1145/3802002.

[2] J. S. Barbosa et al. "A Cost-Effective LLM-based Approach to Identify Wildlife Trafficking in Online Marketplaces." *Proceedings of the ACM on Management of Data*, 2025. doi:10.1145/3725256.

[3] Sentence Transformers. "all-MiniLM-L6-v2" model card. Hugging Face model documentation.

[4] M. Lesci and A. Vlachos. "AnchorAL: Computationally Efficient Active Learning for Large and Imbalanced Datasets. *NAACL*, 2024. https://aclanthology.org/2024.naacl-long.467/.

[5] C. Coleman et al. "Similarity Search for Efficient Active Learning and Search of Rare Concepts (SEALS)." *AAAI*, 2022. https://arxiv.org/abs/2007.00077.

[6] G. Lesci and A. Vlachos. "AnchorAL: Computationally Efficient Active Learning for Large and Imbalanced Datasets." *NAACL*, 2024. https://aclanthology.org/2024.naacl-long.467/.